



High-temperature absorption line shape parameters for CO₂ in the 6800–7000 cm⁻¹ region from dual frequency comb measurements up to 1000 K

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ABSTRACT

We present the first systematic study of self-broadened CO₂ line shape parameters in the 00031 ← 00001 and 01131 ← 01101 bands at high temperature. We report temperature-dependent broadening coefficients and pressure shift coefficients for more than 40 transitions in the 00031 ← 00001 band, and 75 transitions in the 01131 ← 01101 hot band. The majority of these parameters have never been measured at high temperature. The new parameters are determined from 12 spectra measured with a broadband, high-resolution dual frequency comb absorption spectrometer in a high-temperature gas cell from room temperature to 980 K. We fit the measured spectra using a multispectrum fitting approach to derive line shape parameters with both a Voigt and speed-dependent Voigt profile. We compare the measured parameters to calculated line shape parameters as well as to the new parameter set developed for the HITRAN2020 database. The new measurements demonstrate strong agreement with HITRAN2020 (within 1% in some cases), and provide new measurements of the temperature dependence of the broadening and pressure shift coefficient that will improve the accuracy of the database at high temperature. We extend our results to high $|m|$ transitions using a set of empirical models for the broadening and pressure shift coefficients, which we fit to our data and validate for high-temperature transitions up to $|m| = 98$. We use these models to construct a set of database files that can be used to supplement the HITRAN database for self-broadened CO₂ in the 6800 – 7000 cm⁻¹ region. These new parameters significantly improve the accuracy of absorption models for high-temperature CO₂ in this frequency range, and also provide useful data to test and improve methods used to calculate CO₂ line shape parameters.

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1. Introduction

Accurately modeling the spectrum of carbon dioxide at high temperature is a central problem in fields ranging from planetary science to combustion. In planetary science, temperature-dependent absorption models are used to determine radiative transfer in exotic planetary atmospheres and to aid in interpreting observations (e.g. to differentiate and identify absorption by trace gases) [1,2]. In combustion, CO₂ is a common target for laser-based sensors that derive gas properties from absorption spectra measured in combustion systems [3]. For both applications, self-broadened CO₂ absorption models become increasingly important due to the large fraction of CO₂ in combustion processes and many planetary atmospheres (e.g. Venus, 96.5% [4] and Mars, 95% [5]).

Accurate absorption models for carbon dioxide require knowledge of the position (σ_0) and intensity (S) of CO₂ quantum state transitions. In addition, they require a set of temperature-dependent and species-specific line shape parameters such as the broadening coefficient (γ) and the pressure shift coefficient (δ) that describe the shape of each transition and its evolution with pressure and temperature. These parameters are stored in published spectral databases such as HITRAN and HITEMP [6,7]. For CO₂, recent HITRAN versions primarily use calculated spectral parameters, including theoretical or semi-empirical line positions and intensities [8,9] as well as line shape parameters calculated using the models described in Refs. [10–13] and [14]. These calculations have the benefit of providing a consistent set of spectral parameters to model all known transitions, including a very large number of weak, high-temperature transitions that are difficult to measure in the laboratory.

However, despite the benefits and emphasis put on calculated spectral parameters, high quality experimental parameters are still

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required, both as fundamental inputs to many global models for line shape parameters and as a means of testing and validating the calculations across a range of conditions [10,14]. These validations are particularly important at high temperatures, where spectral models must increasingly rely on accurate, calculated parameters to account for the large number of high J'' transitions and hot bands that are only observed at very high temperatures. Furthermore, many models for line shape parameters do not include beyond-Voigt effects (e.g. temperature dependence of self-shift, speed dependence, Dicke narrowing, etc.) that can significantly improve agreement between measured and modeled absorption features [e.g. 15–17]. In this case, parameters derived from experiments are often the only source to model non-Voigt effects in measured spectra. Despite this importance, only a small fraction of the past studies of self-broadened CO_2 include non-Voigt effects, and many bands have never been measured at high temperature [18].

Here, we report measurements of self-broadened CO_2 line shape parameters and their temperature dependences for transitions in the $00031 \leftarrow 00001$ and $01131 \leftarrow 01101$ bands. Accurate line shape parameters for the $00031 \leftarrow 00001$ band are of particular interest due to its relatively strong absorption near the $\sim 1.55 \mu\text{m}$ telecommunication band commonly used for laboratory spectroscopy and sensing. Despite these benefits, line shape parameters for these bands have only been measured in four prior studies [19–22], with temperature dependence only published recently for eight transitions in the $00031 \leftarrow 00001$ band [21].

In this study, we measure broadening coefficients, pressure shift coefficients, and their temperature dependences for more than 40 transitions in the $00031 \leftarrow 00001$ band and 75 transitions in the $01131 \leftarrow 01101$ hot band with both a Voigt and speed-dependent Voigt profile. As mentioned above, the majority of these transitions have never been measured at high temperature. The new parameters are determined from multispectrum fits to spectra measured with a broadband, high-resolution dual frequency comb absorption spectrometer at temperatures up to 980 K. We compare the new parameters to values calculated using different models for CO_2 line shape parameters [10,14], as well as to the updated parameter set developed for HITRAN2020 [23]. We also fit the measured line shape parameters with empirical models for the broadening coefficient and the pressure shift coefficient, which we use to extend our results to transitions we could not directly measure. We validate the empirical models for high-temperature transitions up to $|m| = 98$. Taken as a whole, these results significantly improve self-broadened CO_2 absorption models in the $6800\text{--}7000 \text{ cm}^{-1}$ region while also providing comparisons that can be used to validate and improve models for CO_2 line shape parameters at high temperatures.

2. Experimental details and data preparation

The spectra used in this study were measured in a high-uniformity, high-temperature gas cell at the University of Colorado Boulder. A full description and characterization of the specialized gas cell is the subject of a dedicated paper [24], and here we only describe the most relevant details. The gas cell consists of a $45.82^{+0.11}_{-0.34} \text{ cm}$ quartz cell housed at the core of a high-temperature ceramic tube furnace. The quartz cell is surrounded by a molybdenum tube that acts as a heat spreader and increases the temperature uniformity of the sample gas. The temperature distribution inside the furnace is monitored by seven K-type thermocouples, including two in situ thermocouples that directly monitor the temperature of the sample gas near the cell windows. The temperature uniformity across the absorbing region (defined as the largest percent deviation from the mean temperature) is better than 2% for all conditions studied in this paper. The sample gas pressure is mon-

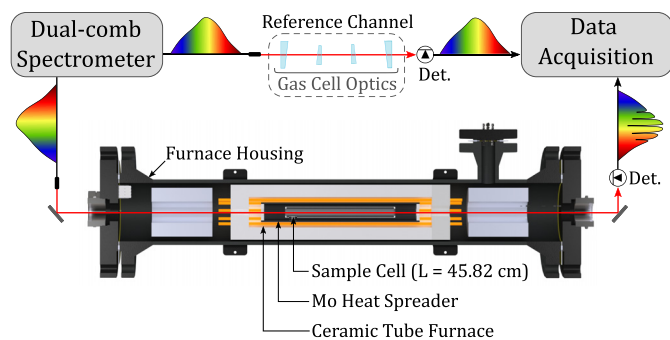


Fig. 1. Overview of the experimental configuration used to measure the high-temperature CO_2 spectra. Light from a dual-comb spectrometer is passed through a CO_2 -filled, quartz sample cell housed at the core of a high-temperature ceramic tube furnace. Light from the spectrometer is simultaneously passed through an additional external reference channel to record the non-absorbing baseline intensity of the spectrometer. Further details on this gas cell and experimental configuration are given in Ref. [24].

Table 1

The temperature and pressure conditions of each measured spectrum and their associated uncertainties. The specified temperatures reflect the mean gas temperature, and the uncertainty calculations are discussed further in Appendix 1.

Temperature (K)	Pressure (Torr)
298 ± 2.2	75.2 ± 0.2 ; 151.6 ± 0.4 ; 335.5 ± 0.8
496 ± 6.4	75.0 ± 0.2 ; 154.6 ± 0.4 ; 344.2 ± 0.9 ; 628 ± 1.6
737 ± 9.6	153.9 ± 0.4 ; 336.8 ± 0.8 ; 636 ± 1.6
980 ± 12.7	336.9 ± 0.8 ; 637 ± 1.6

itored using a calibrated capacitance manometer (MKS Baratron 626C) with an accuracy of 0.25% of the reading. An overview of the experimental configuration is shown in Fig. 1, and further details on the temperature, pressure and path length uncertainties associated with the high-temperature gas cell are given in Appendix 1.

We record absorption spectra through the high-temperature gas cell using a broadband, high-resolution, fiber mode-locked dual frequency comb spectrometer (DCS) [25–27]. In dual-comb spectroscopy, two frequency combs with slightly different comb tooth spacing are combined and passed through an absorbing sample. Interference of the two frequency combs maps absorption information sampled by individual pairs of comb teeth at optical frequencies to heterodyne beat frequencies in the radiofrequency range that can be measured on a single photodetector [28]. This approach combines the unique benefits of mode-locked frequency combs (e.g. broad bandwidth and frequency stability) with the high resolution afforded by the comb-tooth-resolved detection scheme.

The specific DCS system used for these measurements [25–27,29] is based on two erbium fiber mode-locked frequency combs with comb tooth spacing of $\sim 0.0066 \text{ cm}^{-1}$ and a nominal bandwidth of $\sim 5500\text{--}8000 \text{ cm}^{-1}$, which we optically filter for this study to span $6800\text{--}7000 \text{ cm}^{-1}$. The two self-referenced frequency combs are stabilized to a common, narrow-linewidth continuous wave (CW) reference laser. A simple feedback to the temperature of the reference laser ensures that the pulse repetition rate of each comb (which sets the comb tooth spacing) remains within $\pm 5 \text{ Hz}$ of its expected value. All measurements of the lock frequencies and repetition rates are referenced to a GPS-disciplined oscillator (Jackson Labs Fury) with stability better than 10^{-11} . We estimate the absolute frequency accuracy of the DCS system to be $\sim 1.7 \times 10^{-4} \text{ cm}^{-1}$ based on the stability of the CW reference laser. Further details of this uncertainty estimate are given in Appendix 1.

We use the high-temperature gas cell and DCS to measure spectra across the range of pressure and temperature conditions summarized in Table 1. All spectra are collected for samples of pure

CO₂ (AirGas Instrument Grade, >99.99% purity). We measure the absorption of the principle CO₂ isotopologue, ¹²C¹⁶O₂. The isotopologue abundance of the sample was independently measured at 98.4665 ± 0.0002% using an isotope ratio mass spectrometer at the CU Boulder Stable Isotopes Laboratory.

We record spectra at each condition using a coherent averaging technique [30] for 60 to 90 min to reach typical RMS absorbance noise of <3 × 10⁻⁴. To account for the variable intensity spectrum of the frequency comb lasers (which form the non-absorbing 'baseline'), we normalize each measured spectrum using a background spectrum measured while the gas cell is under vacuum. We then further normalize each spectrum using a second background spectrum measured concurrently with the sample spectrum that corrects for temporal intensity drift in the spectrometer. In the latter spectrum, light from the spectrometer is measured through an identical set of windows to those used on the gas cell, but on a second laser path that does not pass through the gas cell itself (see Fig. 1). In practice, small, broadband residual baseline variations remains after this normalization process (<5%). To account for the remaining variation, we fit each spectrum with a Chebyshev polynomial (15 coefficients across the entire bandwidth) during the fitting process to extract line shape parameters. Finally, although all optical paths are purged with dry nitrogen during measurements, a small amount of interfering water vapor absorption is still observed in some spectra. This residual water absorption is assumed to be from water present in the bath gas (argon) surrounding the quartz sample cell. We fit and subtract the residual water vapor lines in the impacted spectra using a quadratic speed-dependent Voigt profile and argon-broadened spectral parameters for water vapor [31]. Appendix 1 discusses the uncertainties associated with baseline correction and the removal of interfering water lines.

3. Line shape model and multispectrum fitting

3.1. Line shape model

We fit the measured spectra using the multispectrum fitting code originally proposed in Ref. [32] and used in many past studies to accurately determine line shape parameters from high-resolution spectra [e.g. 16,19,33,34]. In the multispectrum fitting approach, all spectra are fit together to extract the line positions, intensities and line shape parameters that best fit the measured absorption features across the full matrix of temperature and pressure conditions. This method ensures that the retrieved parameters are self-consistent across the full range of experimental conditions while also minimizing possible error sources due to the non-absorbing baseline or correlations between different parameters [16,32,35].

The multispectrum fitting approach generally employs a quadratic speed-dependent Rautian profile formulated as

$F(x, y, a_w, H)$

$$= \frac{2}{\pi^{\frac{3}{2}}} \int_{-\infty}^{\infty} v e^{-v^2} \arctan \left[\frac{x(T) + v}{y(T)(1 + a_w(v^2 - \frac{3}{2})) + H} \right] dv \quad (1)$$

where the molecular speed v is normalized by the most probable speed, and the integral is computed from $v = -4$ to $v = 4$ in 16 steps at each frequency [33,36]. The line position and pressure shift are contained in the temperature-dependent parameter $x(T)$, which for a pure gas is given by $[\sigma - \sigma_0 - P\delta_s(T)]/\gamma_D(T)$. Here, σ is the frequency of the calculation, σ_0 is the 'zero pressure' line center, P is the pressure, $\delta_s(T)$ is the temperature-dependent self-pressure shift coefficient, and $\gamma_D(T)$ is the Doppler width. Similarly, the parameter $y(T)$ concerns the collisional width, and is given (again for a pure gas) by $P\gamma(T)/\gamma_D(T)$ where $\gamma(T)$ is

the temperature-dependent self-broadening coefficient. The speed-dependence parameter (a_w) is equal to the ratio of speed-dependent broadening to the collisional (Lorentz) broadening, and the parameter H concerns the effect of velocity changing collisions (Dicke narrowing) on the line width [33]. To implement Eq. (1) for gas mixtures, the self-broadening and -shift coefficients in the above equations are added linearly with foreign-broadening and -shift coefficients to account for collisions with other gas species in the mixture [33].

This work uses both the Voigt profile (VP) and the quadratic speed-dependent Voigt profile (qSDVP). We obtain the Voigt profile from Eq. (1) in the limit that both speed-dependence and Dicke narrowing effects are set to zero ($a_w = H = 0$). It is well known that the Voigt profile is not a fully accurate model for observed absorption features [37–39]. Line shape parameters determined using a Voigt profile can be biased by the fact that the profile omits several physical processes that can impact the line shape (e.g. speed dependence and Dicke narrowing) [37]. As such, line shape parameters determined using the Voigt profile should be regarded as "effective" parameters, which are most accurate near the conditions in which they were determined. Because the parameters are approximate, several past studies have shown that their impact on the absorption model can change depending on the conditions of the measurement (temperature, pressure, optical depth, etc.) [e.g. 40–42].

In this work, we address the limitations of the Voigt profile by also implementing the quadratic speed-dependent Voigt profile, which we derive from Eq. (1) in the limit that only $H = 0$. The qSDVP improves upon the Voigt profile by considering the effect of molecular speed on the collisional width. Briefly, speed dependence accounts for the fact that relaxation rates are higher (and therefore transition linewidths are broader) for molecules with larger relative speeds [43]. Because the Maxwell-Boltzmann distribution is skewed towards lower speeds, it follows that the speed-dependent width is slightly narrower than that of a Voigt profile that only considers the mean relative velocity [38,39,44]. Speed dependence is different from Dicke narrowing, which arises due to the fact that collisions can reduce the average molecular speed, thereby reducing the Doppler component of the line width [45]. We refer the reader to other sources (e.g. [37–39,41,44,46,47] and references therein) for a more detailed description of these effects.

We adapt Eq. (1) to model high-temperature absorption features by adopting temperature-dependence models for the broadening coefficient and pressure shift coefficient. For the broadening coefficient, we adopt the widely used power-law temperature dependence [48]

$$\gamma(T) = \gamma(T_{ref}) \left(\frac{T_{ref}}{T} \right)^{n_\gamma} \quad (2)$$

where we select $T_{ref} = 296$ K and the temperature dependence is captured for each measured transition in the exponent n_γ .

There is less of a consensus on an appropriate temperature-dependence model for the pressure shift coefficient, which is largely driven by the scarcity of pressure shift measurements at different temperatures [48]. As such, we model the temperature dependence of the pressure shift coefficient using a power-law expression similar to Eq. (2) but with exponent n_δ . We note that a double power law (DPL) expression was recently introduced to more accurately model the temperature dependence of the broadening coefficient and pressure shift coefficient over wide temperature ranges [48]. We do not use the DPL expression for this work because we do not detect any deviations from the single power law model (Eq. (2)) above the noise level for the spectra and temperature range used in this study. We test the single power law model for the temperature range of this study in more detail in Appendix 2.

In addition to the temperature dependence of the broadening and shift coefficients, our multispectrum fitting code makes the assumption that the speed-dependence parameter (a_w) is independent of temperature. There remains considerable uncertainty about the temperature dependence of the speed-dependence parameter [49], with some studies finding that the parameter is theoretically independent of temperature [47,50], while others finding that it has a unique temperature dependence [14,51]. We proceed with the assumption that the speed-dependence parameter is temperature independent following other work that has employed the same multispectrum fitting code [16,34–36]. Again, we assess this assumption in more detail in Appendix 2.

Finally, we note that our spectral model does not include the effect of line mixing, which is known to be significant in CO₂ even near atmospheric conditions [14–16,35]. In Appendix 3, we estimate the effect of omitting line mixing on the retrieved line shape parameters. We find that the omission of line mixing in our spectral model changes the retrieved broadening coefficients by an average of ~0.39% and the pressure shift coefficients by ~3.8%. A full investigation of line mixing and its temperature dependence will be the subject of future work, which will benefit from the line-by-line database of temperature-dependent line shape parameters developed in this study.

3.2. Multispectrum fitting approach

We structure our specific multispectrum fitting approach to favor the retrieval of temperature-dependent line shape parameters over individual line centers and intensities. As such, we do not fit the line center and intensity individually on a line-by-line basis. Instead, we fit a single global parameter Δ_S that scales the intensity of all transitions in a given vibrational band relative to their database values. Similarly, we fit a single global parameter Δ_σ that serves as an offset from database line centers. With this format, the line position and intensity of each line i can be expressed as

$$\sigma_i = \sigma_{i,Database} + \Delta_\sigma (v' \leftarrow v'') \quad (3)$$

$$S_i = S_{i,Database} \times \Delta_S (v' \leftarrow v'') \quad (4)$$

where σ denotes the line center, S denotes the intensity, and $v' \leftarrow v''$ indicates that values for Δ_S and Δ_σ are determined separately for each vibrational band included in the multispectrum fit. By constraining the intensities and line positions in this way, we force the line positions and intensities in the solution to remain consistent with the quantum mechanical expressions used in calculations [8,9,52,53], while also allowing the fitting process to compensate for experimental uncertainties that may impact the retrieved intensities or line centers. Furthermore, these constraints also mitigate unwanted correlations between the line center and the pressure shift coefficient and the intensity and the broadening coefficient [35]. However, it is important to note that this constraint format cannot compensate for relative (i.e. state-to-state) errors in the database line centers and intensities. As such, sources of reference data must be carefully chosen to avoid propagating relative errors in the line centers and intensities into the measured line shape parameters. We discuss our choice of reference data for this analysis in more detail in Section 4.1.

Before running a multispectrum fit, we also assign a weight to each spectrum based on the procedure described in Refs. [16,34,35]. In this approach, a weight of 1.0 is assigned to the spectrum with the highest signal to noise ratio (SNR). The weight of each subsequent spectrum (which each have a similar, but not identical SNR) is then calculated as the inverse square of the ratio of the highest SNR to the SNR of that spectrum. By weighting in

this way, we help ensure that the retrieved line shape parameters are optimized for all temperature and pressure conditions, and are not biased by the differing SNR of each spectrum included in the fit [35].

After applying the constraints and assigning an appropriate weight for each spectrum, we begin the multispectrum fitting process by incrementally adding lines to the fit solution while varying the line shape parameters for the quadratic speed-dependent Voigt profile and the relevant constraint parameters Δ_S and Δ_σ . We continue to add progressively weaker lines to the fit solution until the retrieved spectral parameters exhibit excessive scatter or very high uncertainties (indicative of unstable or non-physical fit results). After reaching this point when fitting qS-DVP parameters, we repeat the process while fixing the speed dependence at zero to measure effective Voigt parameters. As discussed later in Section 4.1, we do not fit the line center and intensity constraint parameters when using a Voigt profile, and instead fix these parameters at the values determined using the qSDVP.

Following the first complete multispectrum fit, we fit the measured parameters using empirical models for the broadening coefficient, pressure shift coefficient, and their temperature dependence exponents (discussed in detail in Section 5). We use the resulting models to estimate line shape parameters that are not listed in HITRAN (e.g. the self-pressure shift coefficient and its temperature dependence). We then use these preliminary estimates to provide better initial values for the multispectrum fitting code, and repeat the fitting process a second and final time using the more complete initial database. Fig. 2 shows representative fit results at 298 K and 737 K using both the Voigt and qSDVP profile after the final multispectrum fit.

In practice, we find that the total number of lines that we can accurately fit is limited in two ways. First, band heads present in the R-branch of many vibrational bands in this region limit the measurement of individual spectral parameters due to overlap between multiple transitions. Second, we find that our fits are also limited to lines with measurable absorption at 298 K. Although in principle the fitting program should be able to extract accurate parameters for lines that only absorb at high temperatures, we find that the retrieved line shape parameters for these lines exhibit excessive scatter that we judge to be nonphysical. We attribute this limitation to correlations between the line shape parameters and their temperature dependence exponents that become unmanageable when there is no measurable absorption to anchor the parameter at the reference temperature (296 K).

Finally, in an effort to present only the highest-confidence parameter set, we only report parameters with statistical uncertainty <25%, and no excessive residual in the fit result that would indicate an unstable or non-physical fit solution. Furthermore, the majority of the measured parameters exhibit smooth trends with the rotational quantum number that are consistent with theoretical predictions [13]. Similar to Ref. [19], we judge fit results that deviate significantly from this smooth behavior to be non-physical, and we do not report these parameters. We group the measured parameters as those that impact the line width (γ , n_γ and a_w) and those that impact the shift (δ and n_δ). We do not report any of the parameters in a given group if one of the parameters does not satisfy the requirements outlined above. For example, if the speed-dependence parameter of a line exhibits statistical uncertainty >25%, we also do not report the broadening coefficient and its temperature-dependence exponent for that line. Even with this quality control criteria, our final measured parameter set includes values for more than 40 lines in the 00031 $\leftarrow$ 00001 ($3\nu_3$) band, and 77 lines in the 01131 $\leftarrow$ 01101 hot band. Table 2 summarizes the total number of reported parameters extracted using both the Voigt and qSDVP profiles.

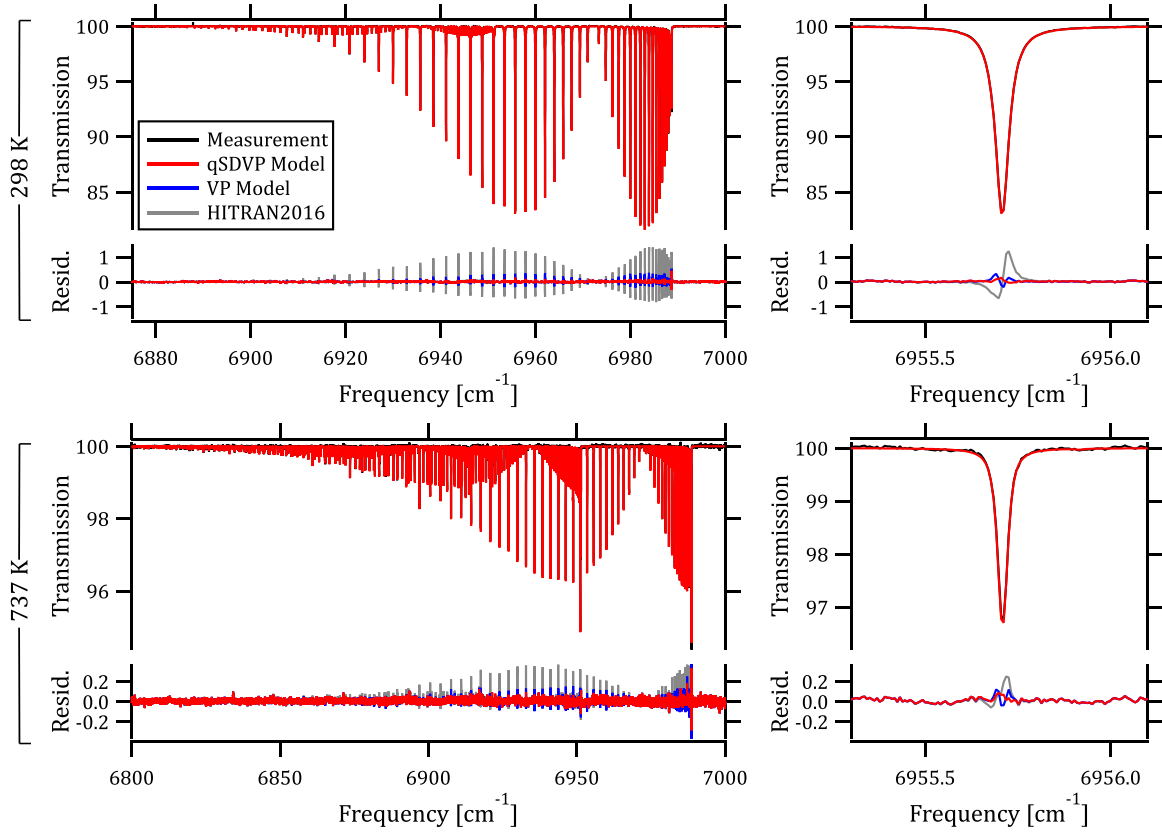


Fig. 2. Spectra measured at ~ 150 torr and 298 K (top) and 737 K (bottom) along with the fit results obtained using both the VP and qSDVP. For clarity, the Voigt profile model is only shown in the residuals. The residuals also show a comparison to models generated with the HITRAN2016 database. The insets on the right hand side show the P18e line of the 00031 $\leftarrow$ 00001 band. The improvement in the fit residual when using the qSDVP is evident.

Table 2

Total number of lines with reported fit parameters for each vibrational band analyzed using the multispectrum fitting approach. Vibrational band notation follows that of Ref. [17].

Band	Number of Lines with Reported Parameters	
	$\gamma, n_{\gamma}, (a_w)$	δ, n_{δ}
00031 $\leftarrow$ 00001 [VP]	46	38
00031 $\leftarrow$ 00001 [qSDVP]	43	41
01131 $\leftarrow$ 01101 [VP]	77	

4. Multispectrum fit results and discussion

4.1. Line positions and intensities

As discussed in the preceding section, we parameterize the intensity and line center using uniform scaling factors and offsets from database values. In practice, we only retrieve values for the line center constraint parameter (Δ_{σ}) for the 00031 $\leftarrow$ 00001 band because the weak absorption in the 01131 $\leftarrow$ 01101 hot band prevents accurate shift retrievals for that band. For the 00031 $\leftarrow$ 00001 band, we measure the line center offset relative to HITRAN2016 using the quadratic speed-dependent Voigt profile. We do not report the line center offset measured using the Voigt profile to avoid any potential biases due to the increased residuals present in the Voigt profile fits. Using the qSDVP, we measure the line center constraint parameter to be $0.59 \pm 1.74 \times 10^{-4} \text{ cm}^{-1}$ for the 00031 $\leftarrow$ 00001 band. We describe our estimate of the uncertainty in the line center offset in Appendix 4. In this case, the estimated uncertainty in the constraint parameter covers zero, meaning that we do not detect a meaningful shift relative to the

HITRAN2016 database. However, we note that our estimated uncertainty ($\pm 1.74 \times 10^{-4} \text{ cm}^{-1}$) is significantly lower than the upper uncertainty bound listed for the HITRAN2016 line positions ($\pm 0.001 \text{ cm}^{-1}$). As such, this measurement suggests that the true uncertainty in the HITRAN line positions for the 00031 $\leftarrow$ 00001 band is much better than the value suggested by the HITRAN database uncertainty codes. Finally, we also note that our results appear to be in excellent agreement with the recent results shown in Fig. 11 of Birk et al. [22], which shows a systematic shift on the order of $0.5 \times 10^{-4} \text{ cm}^{-1}$ relative to HITRAN2016 for the 00031 $\leftarrow$ 00001 band.

We also measure the intensity constraint parameter (Δ_S) based on Eq. (4). For the 00031 $\leftarrow$ 00001 band, we measure Δ_S relative to the intensities calculated by Zak et al. [9], as recent measurements have shown that HITRAN2016 intensities exhibit errors in the relative intensities within the 00031 $\leftarrow$ 00001 band which are not present in the values calculated by Zak et al. [22,54]. Indeed, recent measurements reported by Birk et al. [22] demonstrate that the intensities calculated by Zak et al. accurately replicate the relative intensities for transitions within this band, but must be scaled by a factor of 1.122 in order to reproduce the absolute intensities of measured transitions. Here, we measure the band-wide intensity scaling factor (Δ_S) for the 00031 $\leftarrow$ 00001 band using the qSDVP. For the 00031 $\leftarrow$ 00001 band, we measure the intensity scaling factor (Δ_S) to be 1.12 ± 0.02 . This value is in excellent agreement with the scaling factor measured by Birk et al. Again, we do not report scaling values for this band measured using the Voigt profile, as past studies have shown that the Voigt profile can bias retrieved intensity values [e.g. 55,56].

We also measure the constraint parameter (again relative to the intensities of Zak et al.) for the 01131 $\leftarrow$ 01101 hot band. Here, we

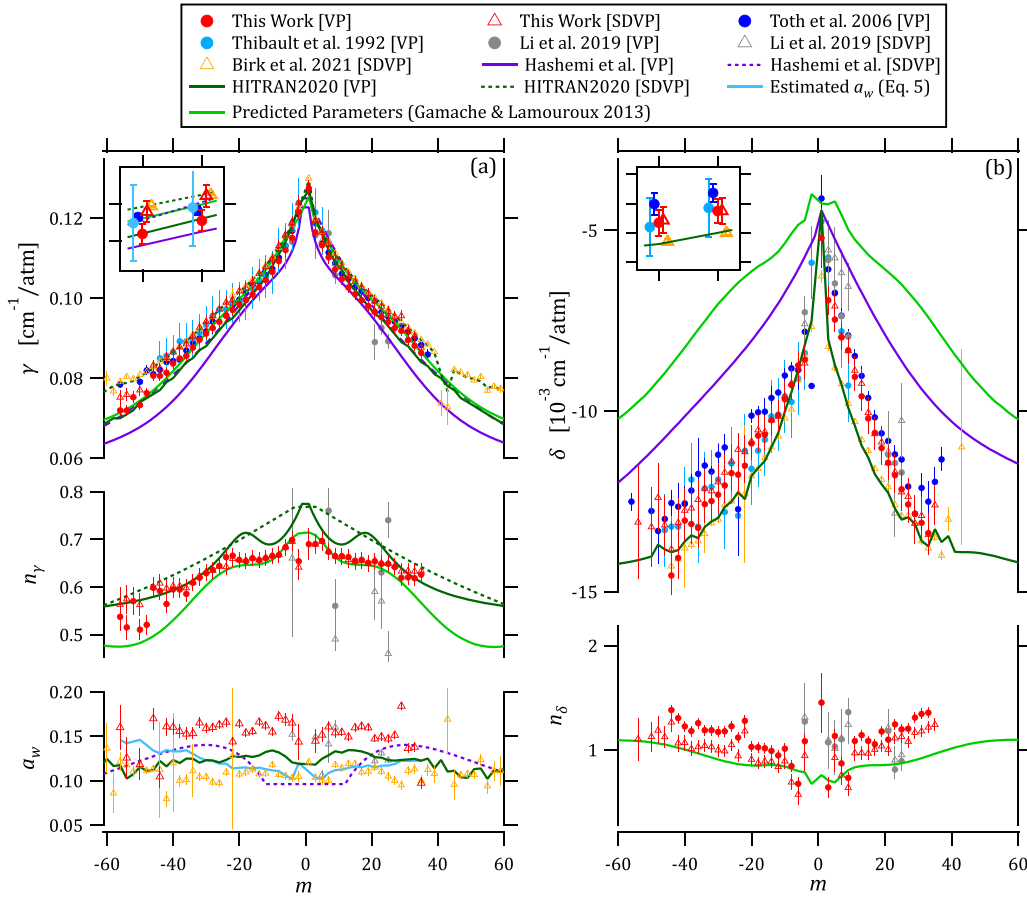


Fig. 3. Panel (a) shows the self-broadening coefficients (γ), temperature-dependence exponents for the self-broadening (n_γ), and speed-dependence parameters (a_w) measured for the 00031 $\leftarrow$ 00001 band with the multispectrum fitting approach. Panel (b) shows the measured pressure shift coefficients (δ) and their temperature dependence exponents (n_δ). Also plotted are values from past measurements [19–22], calculated parameters [10,14], and the parameter set developed for the HITRAN2020 database [23]. Insets highlight the relative agreement between the different data sources for the P14e and P16e transitions.

measure the scaling factor using a Voigt profile, as these transitions are too weak to reliably measure the speed dependence. For this band, we measure the intensity scaling factor to be 1.13 ± 0.02 . Again, this result appears to be consistent with the findings of Birk et al., who note that the relative intensity errors in the HITRAN2016 database discussed above for the 00031 $\leftarrow$ 00001 band also appear to be present for the 01131 $\leftarrow$ 01101 band [22]. We discuss our estimate of the uncertainty in the measured intensity constraint parameters in Appendix 4.

4.2. Line shape parameters and temperature dependences

The primary motivation of this work is the determination of temperature-dependent line shape parameters for bands in the 6800 – 7000 cm^{-1} region. To the best of our knowledge, self-broadened line shape parameters for this spectral region have only been measured in four prior studies [19–22], with temperature-dependent values only published recently for eight lines in the R-branch of the 00031 $\leftarrow$ 00001 band [21].

Fig. 3 shows the broadening coefficients, pressure shift coefficients, speed-dependence parameters, and temperature dependence exponents measured using the multispectrum fitting approach for the 00031 $\leftarrow$ 00001 band. The parameters are plotted against the index m , which is equal to $-J''$ in the P-branch and $J'' + 1$ in the R-branch. The parameters measured in this study are shown in red, with closed circles designating Voigt profile retrievals, and open triangles marking parameters measured with the qSDVP. Error bars represent the total estimated uncer-

tainty in the measured parameters, which we assess in Appendix 4.

In Fig. 3, we also compare the retrieved parameters to past measurements [19–22], calculated parameters [10,14], and to the parameter set developed for the HITRAN2020 database [23]. We compare the broadening and pressure shift coefficients measured with a Voigt profile to the past measurements of Toth et al. [19], Thibault et al. [20], and Li et al. [21]. Specifically, for the Voigt profile retrievals, we measure average percent differences of 1.8%, 2.5% and -1.6% respectively for the broadening coefficient, and -6.6% , -1.5% and -8.2% respectively for the pressure shift coefficient. We note that the direct comparison of parameters measured with a Voigt profile can be nuanced because the measured parameters can be influenced by the specific conditions (e.g. pressure, optical depth, etc.) of the spectra used in the measurement [40–42].

We also compare the broadening and pressure shift coefficients measured using the qSDVP to the coefficients measured recently by Birk et al. [22]. In this case, we find that the broadening coefficients are in excellent agreement, with an average percent difference of 0.55%. The pressure shift coefficients reported by Birk et al. are slightly larger in magnitude than our measurements, however exact agreement is not expected because Birk et al. account for the speed dependence of the pressure shift, which we do not model here.

Li et al. also provide temperature dependence exponents for the broadening coefficient and pressure shift coefficient. The measured temperature dependence exponents for the pressure shift coefficient agree qualitatively with Li et al., however it is difficult to

assess agreement in the temperature dependence of the broadening coefficient due to the significant scatter and uncertainty in the published measurements. We also compare the measured speed-dependence parameters to the past measurements reported by Li et al. and Birk et al. The comparison shows that our measured speed-dependence is in reasonable agreement with the values of Li et al., but is slightly elevated relative to the measurements reported by Birk et al. We discuss possible explanations for this difference below.

We also compare the measured parameters to values calculated using the approaches of Gamache and Lamouroux [10] and Hashemi et al. [14], as well as to the updated parameter set developed for the HITRAN2020 database [23]. In the approach of Gamache and Lamouroux, a predictive algorithm is developed based on a large set of CO₂ line shape parameters calculated using complex Robert-Bonamy formalism [10–13]. This approach gives a means to predict line shape parameters for every CO₂ band by explicitly modeling the vibrational dependence of the broadening coefficient and pressure shift coefficient [10]. In a different approach, Hashemi et al. calculate line shape parameters using a set of empirical models fit to experimental data. Pressure shift coefficients for all bands are calculated using an established empirical model [57], and broadening coefficients are calculated for each transition by assuming (with experimental evidence) that the broadening coefficient does not depend on vibration. The parameter set adopted for the HITRAN2020 database [23] uses broadening coefficients based on the models described in Hashemi et al. [14] and the recent measurements of Birk et al. [22].

Fig. 3(a) compares our measured broadening parameters to the two calculation schemes and HITRAN2020. The broadening coefficients measured with the Voigt profile show good agreement with parameters calculated using the approach of Gamache and Lamouroux (0.9% average difference) as well as the Voigt profile parameters included in the HITRAN2020 database (−0.9% average difference). Our measurements using the qSDVP also demonstrate excellent agreement with the speed-dependent broadening coefficients listed in HITRAN2020, in this case differing by only 0.5% on average.

The comparisons also show that the broadening coefficients calculated using the approach of Hashemi et al. systematically underpredict the measured coefficients by 2–12%. While the ultimate cause of this discrepancy is not known, one possibility is a vibrational dependence to the broadening coefficient. While most studies of CO₂ note little to no observed vibrational dependence, a few have found exception in the specific bands studied in this work [19,22,58]. These bands (00031 ← 00001 and 01131 ← 01101) both exchange three quanta in the ν_3 (asymmetric stretch) mode. In a study of the vibrational dependence of CO₂ half widths, Gamache and Lamouroux conclude that exchanges in the ν_3 mode have a two to four times larger effect than exchanges in the ν_1 and ν_2 modes, respectively [58]. The observed agreement with the calculations of Gamache and Lamouroux, which explicitly account for this vibrational dependence, may suggest that this discrepancy is rooted in vibrational effects. However, it is also important to note that the calculations of Hashemi et al. (which were primarily focused on updating *air*-broadening parameters) are carefully validated against measurements in several other bands (although none that exchange multiple quanta in the ν_3 mode). Insofar as these comparisons show excellent agreement, we feel that the observed discrepancies are likely confined only to the bands measured in this study.

Fig. 3(a) also compares the measured and calculated temperature dependence exponents for the self-broadening coefficients. The measured temperature dependence exponents agree well with the calculations of Gamache and Lamouroux up to $m \approx |26|$. Over

this range, the values adopted for HITRAN2020 (which use the values calculated by Hashemi et al. [14]) slightly overpredict the measured temperature dependence. We observe only a minor increase in the temperature dependence exponents near $|m| \approx 26$, which is much less significant than that predicted by Hashemi et al. For higher values of $|m|$, the calculations of Gamache and Lamouroux begin to underpredict the measured values, while the HITRAN2020 values agree well with the measured temperature dependence exponents out to the maximum value of $|m| = 56$.

The measured speed-dependence parameters, as expected, are nearly constant across the band. However, the measured values are slightly inflated relative to the calculations of Hashemi et al. and the values reported in HITRAN2020. We investigate these differences using two methods. First, we assess the speed-dependence retrievals by comparing the broadening coefficients retrieved using the qSDVP and the Voigt profile. We expect the broadening coefficients measured with the VP to be lower than those measured with the qSDVP because the qSDVP broadening coefficients are inflated in order to compensate for the narrowing effect of speed dependence. In our measurements, we find that the broadening coefficients measured with the qSDVP are larger by $\sim 3.2\%$ on average compared with those measured with the Voigt profile. Interestingly, this value agrees fairly well with the parameters developed for HITRAN2020 [23], where the broadening coefficients differ by $\sim 4.5\%$ over the same range of transitions. In a second test, we estimate the speed-dependence parameter using the relation [47,50]

$$a_w = \frac{2}{3} \frac{\alpha}{\alpha + 1} \frac{T}{\Gamma(T)} \frac{d\Gamma}{dT} \quad (5)$$

where T denotes the temperature, α is the perturber-to-absorber mass ratio (equal to one for a pure gas), and Γ is the velocity-averaged halfwidth. Assuming the pressure broadening coefficient follows a single power-law temperature dependence (Eq. (2)), the expression in Eq. (5) evaluates to $a_w = \frac{1-n_\gamma}{3}$ for our case of pure CO₂ [50]. We estimate the speed-dependence parameter using this expression and our measured values for n_γ , and find good agreement with the past measurements and calculations (see Fig. 3(a)). These tests, combined with the agreement between our measured broadening coefficients and the literature values, imply that our speed dependence parameters may be larger in order to compensate for other narrowing mechanisms (e.g. Dicke narrowing) that could impact our measured spectra. Notably, in Appendix 2 we also discuss additional analysis of the speed-dependence parameter that indicates that our fit results are not significantly impacted by our assumption that the speed dependence does not depend on temperature.

Fig. 3(b) provides similar comparisons between the measured and calculated pressure shift coefficients. We find that both calculation routines significantly underpredict the magnitude of the pressure shift coefficient in the 00031 ← 00001 band. The underprediction in the calculations of Gamache and Lamouroux is due to a known issue with the experimental pressure shift data used to train the model [59]. The underprediction in the calculations of Hashemi et al. may be caused by the fact that the empirical shift model is imperfect in its parameterization of the vibrational dependence of the pressure shift coefficient. Again, it is important to note that the empirical shift model used by Hashemi et al. was compared to data in several other vibrational bands, and found to provide much better agreement [14,57]. The pressure shift coefficients listed in the HITRAN2020 database (which are based on the measurements of Birk et al. [22]) are slightly larger in magnitude than our measured shifts. Again, we note that this difference is likely due to the fact that Birk et al. account for the speed dependence of the pressure shift. We also compare the measured temperature dependence exponents for the pressure shift coefficients

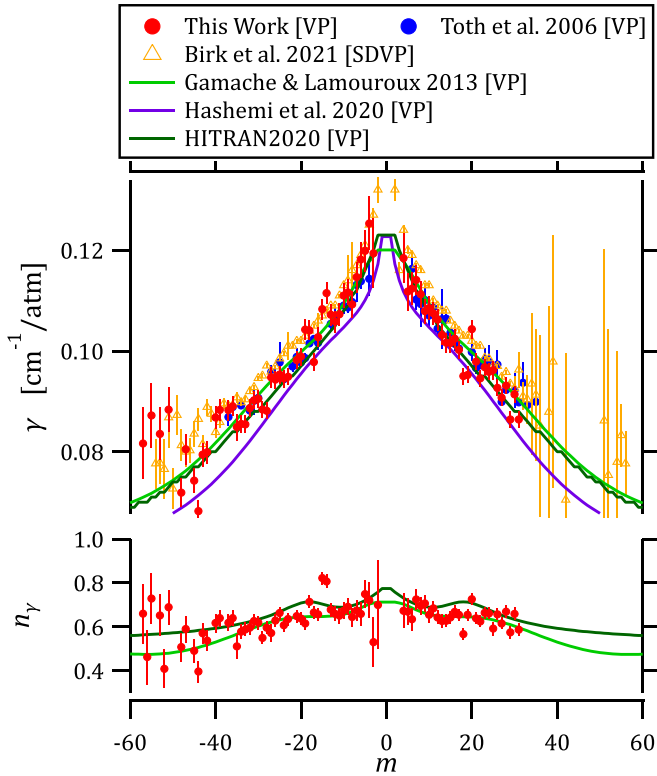


Fig. 4. Measured broadening coefficients (γ) and temperature dependence exponents (n_γ) for the 01131 $\leftarrow$ 01101 band. The measurements are compared to past measurements, calculations, and the updated parameter set for the HITRAN2020 database [10,14,19–22]. Note that for clarity, we omit five measured R-branch transitions from the results of Birk et al. with tabulated uncertainties $>50\%$.

to the calculations of Gamache and Lamouroux. While the comparison shows reasonable agreement, in general we find that the calculated temperature dependence exponents underpredict the measured parameters.

Finally, we also retrieve temperature-dependent broadening coefficients for the 01131 $\leftarrow$ 01101 hot band. Due to the weak nature of this band, the parameters are only measured using a Voigt profile and display more scatter than those in the 00031 $\leftarrow$ 00001 band. We also fit the pressure shift coefficients for these lines, but do not report the values because the residuals exhibit high scatter and uncertainty due to the limited SNR. Fig. 4 shows the measured broadening coefficients and temperature dependence exponents. In general, the comparisons to both past measurements and calculations exhibit the same trends as discussed above. The measured broadening coefficients agree well with the past measurements of Toth et al. [19] and the calculations of Gamache and Lamouroux [10], in both cases differing by $<0.1\%$ on average. Additionally, we find that our measurement agree with the updated broadening coefficients listed in the HITRAN2020 database (differing by -1.9%) [23], and we observe a similar underprediction in the calculated parameters of Hashemi et al. [14]. The measured temperature dependence exponents, although scattered, show reasonable agreement with both calculations. We note that if these comparisons are indeed influenced by vibrational effects, the similarity between these comparisons and those discussed above for the 00031 $\leftarrow$ 00001 band are expected because both bands exchange the same vibrational quanta.

5. Empirical models for collisional parameters

We fit the spectral parameters measured using the multispectrum fitting approach with empirical models for the broadening

coefficient and pressure shift coefficient. These models are useful to provide estimated line shape parameters for lines that could not be fit (e.g. in band heads) as well as a means to estimate parameters for high J'' lines in the absence of better data. The models we use for this purpose are of the same form as those that were independently employed by Hashemi et al. [14] for use in the HITRAN2020 database.

We model the broadening coefficients and their temperature dependence as a function of m using a Padé approximant [14,60,61]. Following [60,61], we choose a 4,4-Padé approximant of the form

$$\gamma(|m|) = \frac{\sum_{i=0}^4 a_i |m|^i}{1 + \sum_{j=1}^4 b_j |m|^j} \quad (6)$$

where a_i and b_j are coefficients fit to the data. Fig. 5 shows the Padé approximants fit to the broadening coefficients and their temperature dependence exponents for both the Voigt and qSDVP retrievals. In both cases, the functions fit the measured parameters well, with differences generally within 1% for the broadening coefficients, and 2% for the temperature dependence exponents. In addition, we will show in Section 6 that the fit results match well with individual high-temperature fits of the broadening coefficients out to $|m| \approx 90$ and beyond. Table 3 lists the coefficients for the Padé approximants that are fit to the data, and which are then used to model the broadening coefficient and the temperature-dependence exponent as a function of the rotational quantum number.

To model the pressure shift coefficients, we adopt the empirical model proposed by Hartmann [57]. This model has been fully summarized in [14,57,62], and here we list only the essential details for completeness. In this model, the pressure shift coefficient for a rovibrational transition $(v'_1, v'_2, v'_3, J') \leftarrow (v''_1, v''_2, v''_3, J'')$ is expressed as [57]

$$\delta[(v'_1, v'_2, v'_3, J') \leftarrow (v''_1, v''_2, v''_3, J'')] = (J'' - J') \delta_{\text{rot}}(|m|) + (y_1 \Delta v_1 + y_2 \Delta v_2 + y_3 \Delta v_3) \delta_{\text{vib}}(|m|) \quad (7)$$

where $\Delta v = (v'_i - v''_i)$. The function δ_{vib} is a symmetric function of m that depends on vibration, while δ_{rot} is an antisymmetric function of m that is independent of vibration. As such, δ_{rot} is constructed from measured pressure shift coefficients in the P and R-branches as

$$\delta_{\text{rot}}(|m|) = \frac{\delta_{\text{meas}}[P(-|m|)] - \delta_{\text{meas}}[R(+|m|)]}{2} \quad (8)$$

while δ_{vib} is given (within a scaling factor) by

$$\delta_{\text{vib}}(|m|) \propto \frac{\delta_{\text{meas}}[P(-|m|)] + \delta_{\text{meas}}[R(+|m|)]}{2} \quad (9)$$

After constructing δ_{vib} and δ_{rot} from experimental data, the two functions are fit with empirical expressions in order to calculate their values for all lines in the band. Following [57], we fit δ_{rot} and δ_{vib} using the empirical expressions

$$\delta_{\text{rot}}(|m|) = \alpha_1^R + \alpha_2^R e^{-|m|/\beta_2^R} + \alpha_3^R e^{-|m|/\beta_3^R} \quad (10)$$

$$\delta_{\text{vib}}(|m|) = \alpha_1^V + \alpha_2^V e^{-|m|/\beta_2^V} + \alpha_3^V e^{-|m|/\beta_3^V} \quad (11)$$

where the α_i and β_i terms are fit coefficients. With satisfactory expressions for both δ_{rot} and δ_{vib} , we fit measured shift coefficients in two bands to determine the coefficients y_1 and y_3 (assuming $y_2 = y_1/2$ [57]) that parameterize the vibrational dependence of the pressure shift coefficient.

We fit the shift model to the pressure shift coefficients measured in the 00031 $\leftarrow$ 00001 band with both the Voigt and qSDVP profile. Since this band only involves a change in vibrational quanta

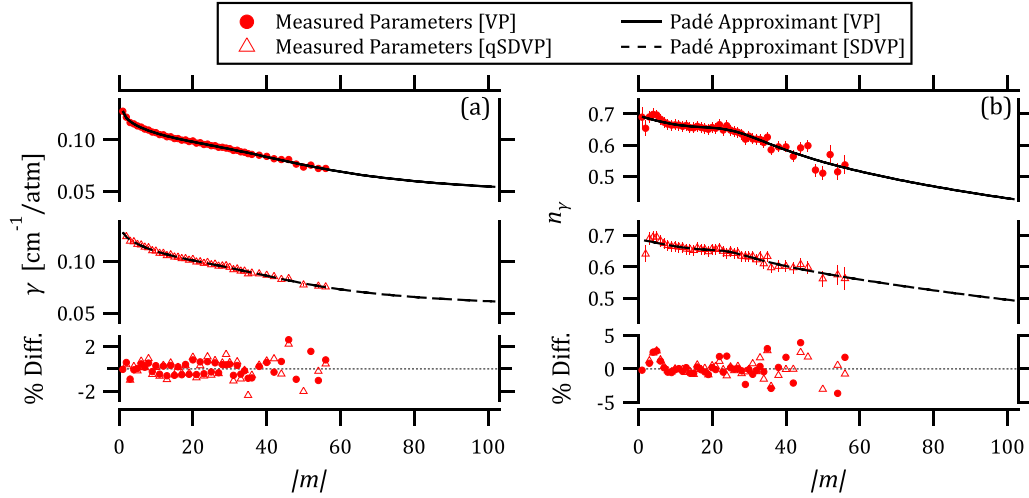


Fig. 5. Padé approximants fit to the broadening coefficients and temperature-dependence exponents measured for the 00031 $\leftarrow$ 00001 band with both the qSDVP and Voigt profile. Panel (a) shows the broadening coefficients fit with 4,4-Padé approximants, while Panel (b) shows similar fit results to the corresponding temperature dependence exponents.

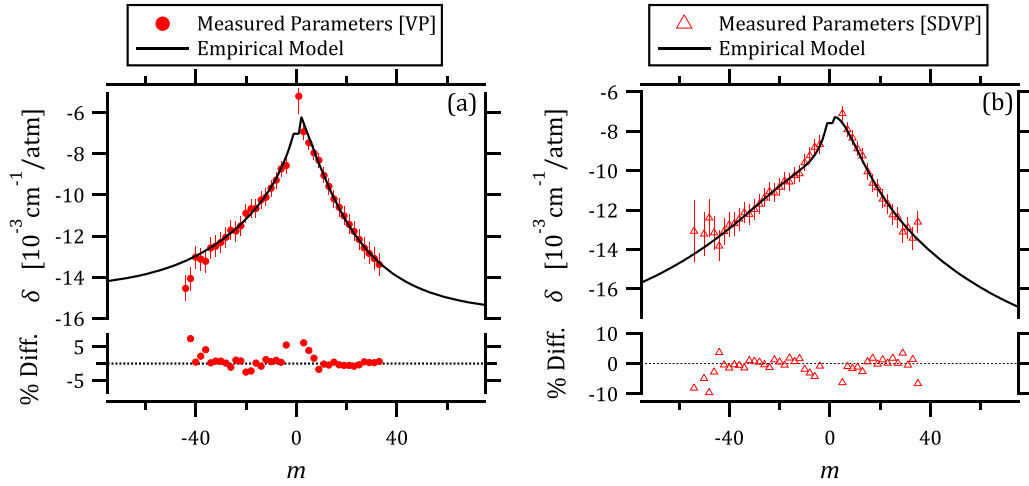


Fig. 6. Results of the empirical pressure shift model [57] fit to the measured shift coefficients. Panel (a) shows the fit to the pressure shift coefficients measured with a Voigt profile, while panel (b) shows the fit to data measured using the qSDVP.

of $\Delta v_3 = 3$, only the y_3 coefficient in Eq. (7) is needed to implement the model. However, in order to complete the model and allow for calculation of shifts in other vibrational bands, we determine the y_1 coefficient based on a fit to published pressure shift coefficients measured in the Fermi triad bands 20011 $\leftarrow$ 00001, 20012 $\leftarrow$ 00001, and 20013 $\leftarrow$ 00001 [19]. Fig. 6(a) and (b) shows the results of the pressure shift model fit to the measured shift co-

efficients in the 00031 $\leftarrow$ 00001 band. For both the Voigt and qSDVP profile, the shift model fits the data to better than 5% for the majority of the lines. The best-fit parameters used to generate the shift models are collected in Table 4.

The measured temperature dependence exponents for the pressure shift are scattered, but exhibit clear structure as a function of $|m|$. We choose to fit the temperature dependence exponents using

Table 3

Best-fit parameters for the Padé approximants used to model the broadening coefficients and temperature dependence exponents measured with the Voigt and qSDVP profiles.

	γ		n_γ		n_δ	
	VP	qSDVP	VP	qSDVP	VP	qSDVP
a_0	5.274×10^1	1.175×10^2	6.943×10^{-1}	6.885×10^{-1}	-3.517×10^2	0.0
a_1	6.601×10^2	2.943×10^3	-4.150×10^{-2}	-5.051×10^{-2}	-1.476×10^2	-1.347×10^2
a_2	-2.399	-3.334×10^1	9.153×10^{-4}	1.086×10^{-3}	-2.963×10^1	3.789×10^1
a_3	0.0	0.0	0.0	0.0	-2.980×10^{-1}	-3.416
a_4	3.357×10^{-3}	2.097×10^{-2}	0.0	0.0	0.0	0.0
b_1	5.497×10^3	2.353×10^4	-5.453×10^{-2}	-6.963×10^{-2}	-3.531×10^2	-9.899×10^1
b_2	7.750×10^1	1.663×10^2	9.402×10^{-4}	1.288×10^{-3}	-1.587×10^1	2.672×10^1
b_3	-2.912	-1.530×10^1	8.293×10^{-6}	6.202×10^{-6}	-2.679×10^{-1}	-2.906
b_4	8.507×10^{-2}	4.482×10^{-1}	0.0	0.0	0.0	0.0

Table 4

Parameters for the best-fit pressure shift model determined from both the Voigt and qSDVP retrievals.

	VP	qSDVP
α_1^R	5.822×10^{-4}	6.174×10^{-4}
α_2^R	-8.831×10^{-2}	-2.694×10^{-1}
β_2^R	1.152×10^{-1}	1.551×10^{-1}
α_3^R	8.723×10^{-2}	2.690×10^{-1}
β_3^R	1.177×10^{-1}	1.570×10^{-1}
α_1^V	-9.201	-9.014
α_2^V	-5.097×10^{-8}	-5.097×10^{-8}
β_2^V	1.949×10^1	1.949×10^1
α_3^V	5.503	5.668
β_3^V	4.195×10^{-2}	1.798×10^{-2}
a_1	1.690×10^{-4}	2.256×10^{-4}
a_3	5.492×10^{-4}	7.209×10^{-4}

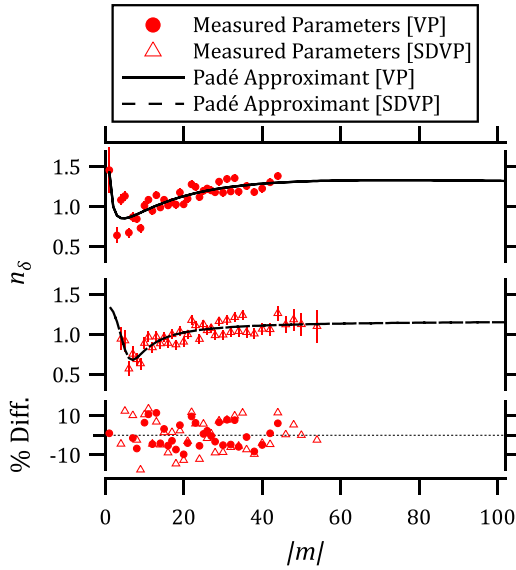


Fig. 7. Best-fit Padé approximants to model the temperature dependence of the pressure shift coefficients.

a Padé approximant of the form described above for the broadening coefficient (Eq. (5)). The fit results are shown in Fig. 7, and the best-fit coefficients are listed in Table 3. Despite the scatter, the empirical model clearly replicates the shape of the temperature-dependence exponents as a function of $|m|$, with differences within 10% for most lines. However, when compared to the Padé approximants fit to the broadening coefficients, the fits in this case are less certain for low- $|m|$ lines due to the increased scatter for these weak transitions. Regardless, similar to the broadening coefficients, we will show in Section 6 that these empirical models are sufficient to replicate pressure shift coefficients measured for high temperature transitions up to $|m| \approx 75$ and beyond.

Finally, it is important to note that the recent work of Hashemi et al. [14] used the same combination of models (i.e. Padé approximants and the shift model of Ref. [57]) to calculate broadening coefficients and pressure shift coefficients for all CO₂ lines in HITRAN. These models were fit to published data in different bands, and were carefully validated against a number of independent measurements. However, we demonstrate in Section 4.2 that these models do not accurately reproduce the broadening coefficients and pressure shift coefficients for the two $\Delta v_3 = 3$ bands measured in this study. As such, given the available information, we feel that the models described in this section are most accurate for the bands studied here, while the models presented in Hashemi et al. should be used outside of the range studied in this paper.

6. Tests of the empirical models for high J lines and hot bands

The multispectrum fit results that form the basis of the models introduced in the preceding section are limited to lines with $|m| \leq 56$ in part due to low absorption in many high $|m|$ transitions at 296 K. Because of this, we extrapolate our empirical models in order to estimate parameters for the high $|m|$ lines beyond the range of the measured parameters. To test and validate the empirical models for these lines, we refit the measured spectra using a different procedure to improve the extraction of parameters for the high $|m|$ transitions visible only in the high-temperature spectra.

Specifically, we refit the spectra grouped by each individual temperature in the test matrix (i.e. the rows of Table 1). In this format, we fit all of the spectra at a given temperature to recover the broadening coefficient and pressure shift coefficient at that temperature (i.e. $\gamma(T)$ and $\delta(T)$). This process is different from the multispectrum approach used previously, where the fit determines the broadening coefficients and pressure shift coefficients at the reference temperature of 296 K along with the values for the temperature dependence exponents to best scale each parameter to the other temperatures in the test matrix.

To implement this “single-temperature” fitting approach, we generally follow the same procedure as outlined in Section 3.2 for the multispectrum analysis. For consistency, we fix the line center and line strength constraints ($\Delta\sigma$ and $\Delta\epsilon$) at the values determined in the multispectrum fit for both the 00031 $\leftarrow$ 00001 and 01131 $\leftarrow$ 01101 bands (see Section 4.1). We then incrementally add lines to the fit while varying the broadening coefficient and the pressure shift coefficient, and fixing the temperature dependence exponents (n_γ and n_δ) at zero. We continue to add progressively weaker lines to the solution until we see that the retrieved parameters consistently exhibit excessive scatter or uncertainty. Finally, we apply the same quality-control criteria as outlined in Section 3.1, but here we include lines with statistical uncertainties up to 50%. We include fit results with higher uncertainty in this analysis because many of the target lines are very weak ($<1\%$ transmission), and have never been measured prior to this study. Ultimately, by applying the single-temperature approach, we improve fits for these weak, high-temperature lines by reducing the number of parameters in the fit solution and eliminating possible correlations between the line shape parameters and their temperature dependence exponents.

Fig. 8 shows the broadening coefficients measured for the 00031 $\leftarrow$ 00001 and 01131 $\leftarrow$ 00001 bands using this approach at each of the four temperatures of the test matrix. As the temperature increases, the state populations increase at higher $|m|$, which we are able to measure using the single-temperature fitting approach. At the maximum temperature of 980 K, we measure broadening coefficients for both bands up to $|m| = 98$, which represents a significant extension over the multispectrum fit results that extend only to $|m| = 56$. The empirical model for the broadening coefficient is overlaid on the measured values at each of the four temperatures. In this case, the empirical model uses both the model for the broadening coefficient and its temperature dependence exponent, which are combined using Eq. (2). The empirical model shows good agreement with the measured parameters for both bands at the four temperatures and over the entire range of rotational quanta. This fact suggests that extrapolation of the empirical models for the broadening coefficient and its temperature dependence provides reasonable estimates for the broadening coefficients up to $|m| = 98$, which we note is nearly the full range of transitions for these bands listed in HITRAN2016 (where $|m|_{\max} = 103$). In addition, this comparison shows that the empirical models developed from fits to data in the 00031 $\leftarrow$ 00001

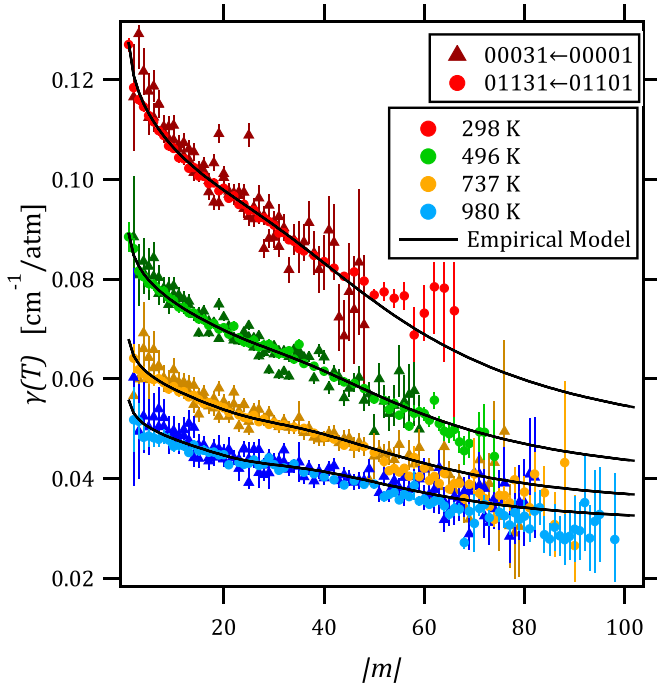


Fig. 8. Broadening coefficients measured with the single-temperature fitting approach at 296 K, 496 K, 737 K, and 980 K. Circular markers denote retrievals in the 00031 $\leftarrow$ 00001 band while darker triangles mark parameters for the 01131 $\leftarrow$ 01101 hot band. The empirical model agrees well with the measured parameters across the full range of temperatures and for transitions up to $|m| = 98$.

band (from Section 5) are also sufficient to estimate parameters in the 01131 $\leftarrow$ 01101 hot band.

In Fig. 9, we show similar data for the pressure shift coefficients measured at the four temperatures. In this case, we only report parameters measured in the 00031 $\leftarrow$ 00001 band because absorption in the 01131 $\leftarrow$ 01101 band is too weak for accurate shift retrievals. We also overlay the empirical shift model developed from Eq. (7) and scaled to different temperatures using the Padé approximant for n_s . Again, the empirical model agrees well with the measured pressure shift coefficients at the four temperatures. Here, the retrieved pressure shift coefficients extend to $|m| \approx 75$, suggesting the empirical shift model is again sufficient to

extrapolate estimated pressure shift coefficients beyond the lines used to develop the model.

In addition to validating the empirical models developed in Section 5, this data may also be useful for testing other line shape parameter models (e.g. the models of Gamache and Lamouroux [10] or Hashemi et al. [14]) for both high temperatures and high $|m|$ lines. Indeed, accurate calculated line shape parameters play a particularly important role for these lines because their weak absorption makes them difficult to quantify in the laboratory yet they can still contribute significantly to observed spectra at the scale of planetary atmospheres or in high pressure gases. Although we do not make these comparisons here, this data is included in the Supplementary Material so that it can be used as a reference for future studies.

7. Database file and availability of data

We consolidate our results into a set of database files for the 6800 – 7000 cm^{-1} range that include updated line shape information based on the measurements and empirical models developed in this study. The parameter files are written in the standard 160 character format that is used in the HITRAN database [63]. Non-Voigt parameters that are not included in the standard 160 character entry (e.g. the self-shift coefficients and the temperature dependences of self-broadening and shift coefficients) are appended to each line and separated by commas. This format is compatible with widely-used simulation tools such as the HITRAN Application Programming Interface (HAPI) [64]. We provide a total of four database files grouped into two sets designed for use with either the Voigt profile or qSDVP. Again, it is important to note that the parameters determined using the Voigt profile are effective parameters, and that the error induced by the Voigt parameters can change depending on the conditions of the measured spectrum (temperature, pressure, optical depth, etc.) [e.g. 40–42]. The general recommendation of the spectroscopic community is to adopt profiles that include narrowing effects [37], and Birk et al. recently demonstrated that optical-depth-dependent errors largely vanish when these narrowing effects are taken into account [40]. As such, we strongly recommend use of the qSDVP database in order to mitigate potential errors due to extrapolation of the effective Voigt parameters. That said, we provide the Voigt profile database to be consistent with HITRAN2020, which simultaneously supports both parameterizations. Further details on the structure of both database files are given below.

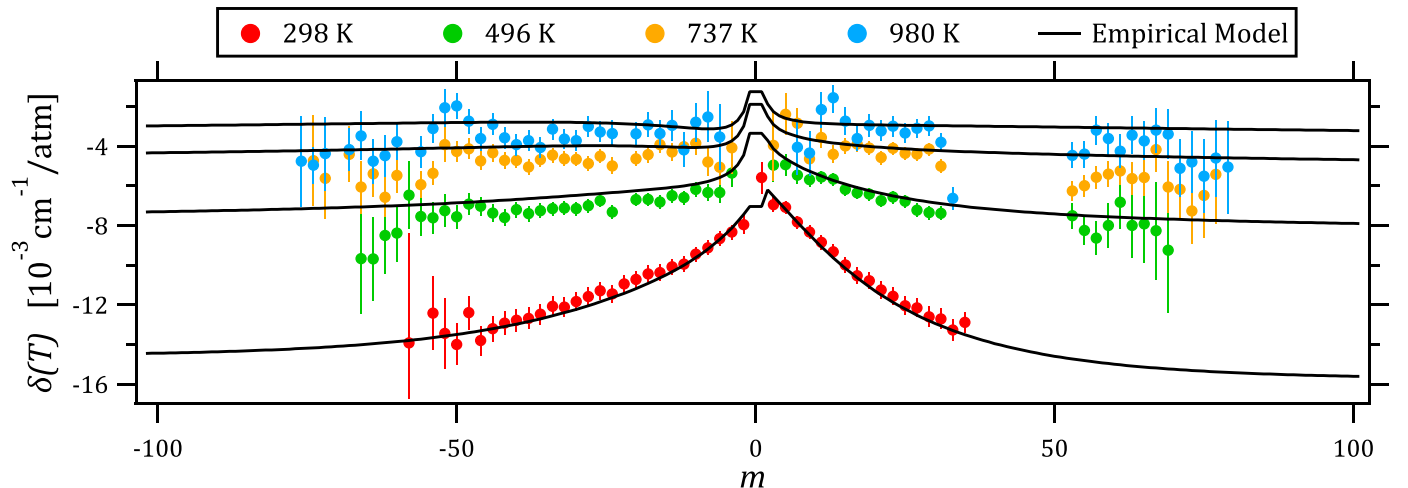


Fig. 9. Pressure shift coefficients measured for the 00031 $\leftarrow$ 00001 band at 296 K, 496 K, 737 K, and 980 K. The empirical model for the pressure shift shows good agreement with the measured shift coefficients up to the maximum temperature of 980 K and for lines up to $|m| \approx 75$.

For each line shape profile, we generate an “empirical” database file, where spectral parameters are generated using the models developed in Section 5. We also construct a “hybrid” file, where measured parameters are substituted for the lines measured in the multispectrum fits. We use two data sources to construct the parameter files, the first being the parameters measured in this work, and the second being the new parameter set developed for HITRAN2020 [14]. In Fig. 8, we show that the empirical models developed in this study for the broadening coefficients in the 00031 $\leftarrow$ 00001 band also accurately model the coefficients measured in the 01131 $\leftarrow$ 01101 band (see Fig. 8). Given this result, we make the assumption that the empirical models developed in this study are the most accurate for modeling the other vibrational bands in this spectral region that have a change in vibrational quanta of $\Delta v_3 = 3$ (and no change in v_1 or v_2). As such, we use the parameters measured in this study to update all bands with $\Delta v_3 = 3$ (nearly all transitions between 6800–7000 cm^{-1}), and we use the spectral parameters specified by Hashemi et al. [14] for all other transitions.

For the 00031 $\leftarrow$ 00001 band, we adopt the line centers listed in HITRAN2016 and Hashemi et al. [6,14], and apply the corresponding offsets based on the measured constraint parameters ($\Delta\sigma$). For the 00031 $\leftarrow$ 00001 and 01131 $\leftarrow$ 01101 bands, we also use the intensities of Zak et al. [9] scaled by the intensity constraint terms reported in Section 4.1. The intensities and line positions of all other bands in the group are fixed at the values listed in HITRAN2016 and Hashemi et al. All self-broadened line shape parameters are adopted from the values determined in this study, either as the measured values (for the “hybrid” database files) or the values calculated from empirical models (for the “empirical” database files). To allow for simulation of non-pure mixtures, we update the air-broadened spectral parameters for each line using the air-broadened widths and shifts determined by Hashemi et al. [14]. As a note, the database files constructed for use with the qSDVP use speed-dependent broadening coefficients (for both air- and self-broadening) that are larger than the broadening coefficients used with a Voigt profile. These database files also include the speed-dependence coefficients for air- and self-broadening. Finally, in addition to the database files, we also include text files of measured spectral parameters in the Supplementary Material.

8. Conclusion

This work represents the first systematic study of self-broadened CO_2 line shape parameters in the 00031 $\leftarrow$ 00001 and 01131 $\leftarrow$ 01101 bands at high temperature. We determine new line shape parameters based on spectra measured with a broadband, high-resolution dual frequency comb absorption spectrometer in a specialized, high-temperature gas cell up to 980 K. We analyze the measured spectra up to $|m| \approx 56$ using a multispectrum fitting approach to extract line shape parameters with both a Voigt and speed-dependent Voigt profile. In total, we report temperature-dependent pressure broadening and shift coefficients for more than 40 transitions in the 00031 $\leftarrow$ 00001 band, and 77 transitions in the 01131 $\leftarrow$ 01101 hot band. The vast majority of these transitions have never been measured at elevated temperature. We compare the measured parameters to calculated values based on the models of Gamache and Lamouroux [10] and Hashemi et al. [14], as well as to the new parameter set developed for the HITRAN2020 database [23]. These comparisons demonstrate strong agreement between the new measurements and the updated HITRAN database (within 1% in some cases), while also providing new measurements of the temperature dependence of the broadening and pressure shift coefficient that will improve the accuracy of the database at high temperature. Furthermore, our results corroborate prior experimental and theoretical studies [10,19,20,22] that show that the broadening coefficients for these bands are sys-

tematically higher than those of many other CO_2 bands. Our hope is that these results and comparisons will be useful to further improve the accuracy of global models for CO_2 line shape parameters.

In addition, we extend our results to high $|m|$ transitions using a set of empirical models for the broadening coefficient and pressure shift coefficient that we fit to our measured parameters. We validate these empirical models against measured parameters for transitions up to $|m| = 98$ in both the 00031 $\leftarrow$ 00001 and 01131 $\leftarrow$ 01101 bands. Through the combination of these empirical models with the measured parameters, we provide database files with updated, self-broadened line shape parameters for CO_2 bands in the 6800 – 7000 cm^{-1} region. These parameters can be used to supplement the HITRAN2020 database in this frequency range and to improve the accuracy of CO_2 absorption models in this spectral region at temperatures relevant to many applications in combustion and planetary science.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendices

1. Description and estimation of systematic experimental uncertainty sources

1.1. Dual frequency comb spectrometer

The dual frequency comb spectrometer (DCS) used in this work can influence the measured line shape parameters in two ways: via instrumental effects that can distort the measure line shapes, and via uncertainty in the spectrometer frequency axis. We discuss these two effects here.

In dual comb spectroscopy, an instrument line shape can be induced in measured spectra if the line width of the individual comb modes approaches or exceeds the spacing between adjacent modes (although this is rarely the case) [28]. In our specific system, we stabilize the repetition rate of each frequency comb using a phase lock to a common CW reference laser. The line width of the CW laser is <2 kHz, and we estimate that the line width of the individual comb modes is then on the order of 10 s of kHz. Since the line width of the individual comb modes is much smaller than the spacing between modes (here ~ 200 MHz), we do not attribute a measurable instrument line shape to our system. That said, dual-comb spectra can also be impacted by other optical and electronic distortions induced by phase noise [65], detector linearity [66], and

cross-phase modulation. We have taken thorough steps to mitigate these effects in our measurements and have carefully tested our spectrometer with measurements of Doppler-limited line shapes. We have not detected any signature of these effects at the conditions studied in this paper.

The frequency accuracy of the dual-comb spectrometer used in this work is fundamentally limited by the accuracy and precision of the reference clock used in the system. This reference clock is used to measure the comb repetition rates as well as the various lock frequencies that are used to stabilize and phase lock each frequency comb. In this work, we reference to a GPS-disciplined oscillator (Jackson Labs Fury). To assess the uncertainty associated with the GPS-disciplined oscillator, we recorded its estimated frequency error (similar to an Allan variance over a 1000 second measurement [67]) before measuring each spectrum in the test matrix. For all the spectra measured in this study, we record an average estimated frequency error of $\sim 1.8 \times 10^{-11}$.

In practice, the true limiting factor in the DCS frequency accuracy is the stabilization of the CW reference laser that is used to control the repetition rate (tooth spacing) of each frequency comb. In this case, we stabilize the CW laser using a simple feedback loop based on the repetition rate of one of the frequency combs. The bandwidth of the feedback loop is ± 5 Hz, meaning the repetition rate can drift 5 Hz (out of ~ 200 MHz) before the feedback loop begins to adjust the CW laser frequency. As such, this feedback loop contributes a worst-case relative frequency uncertainty of $\sim 2.5 \times 10^{-8}$ over the duration of a measurement.

We assume that the uncertainty contribution of the reference clock is negligible relative to the uncertainty due to the CW laser stabilization. As such, we assume the total relative frequency uncertainty of 2.5×10^{-8} , which translates to $\sim 1.74 \times 10^{-4} \text{ cm}^{-1}$ at 6972 cm^{-1} .

1.2. Gas Cell Path Length

We measure the gas cell path length using image analysis with a calibration to convert image pixels to physical length. Our estimate of the uncertainty in the path length includes the following contributions, with their estimated values listed in parentheses:

1. Measurement uncertainty ($\pm 0.09 \text{ cm}$), which we assess as the standard deviation of 11 separate measurements of the path length.
2. Thermal expansion at high temperature ($\pm 0.02 \text{ cm}$).
3. Uncertainty in the wedge angle of the gas cell windows (-0.32 cm).
4. Unknown beam angle through the gas cell ($+0.07 \text{ cm}$).

Based on these uncertainty sources, we estimate the spectroscopic path length of the gas cell to be $45.82^{+0.11}_{-0.34} \text{ cm}$.

1.3. Pressure and temperature measurement

The pressure of each measured spectrum is measured using a calibrated capacitance manometer (MKS Baratron 626C) that is accurate to 0.25% of reading. Additionally, although not reported here, the sample gas pressures are also measured in parallel by two additional capacitance manometers that are optimized for high pressure gases (and therefore have higher uncertainty near atmospheric pressure). Despite the higher uncertainty of the two ancillary sensors, the three readings were found to be in excellent agreement at each condition in the test matrix.

As mentioned in Section 2, the sample gas temperature is monitored by seven, K-type thermocouples, including two immersed in the absorbing gas. All reported gas temperatures reflect the mean measured gas temperatures, which are corrected for radiation effects inside the high-temperature furnace. The temperature uni-

formity (defined as the maximum deviation from the sample gas temperature) ranges from $\sim 5 \text{ K}$ at a gas temperature of 496 K to $\sim 20 \text{ K}$ at a gas temperature of 980 K . The uncertainty in the reported temperatures (Table 1) is the combination of the standard uncertainty for a K-type thermocouple (0.75% of reading) with the uncertainty associated with the radiation correction. We refer the reader to Ref. [24] for a detailed discussion of the gas cell temperature uniformity and the radiation correction.

1.4. Isotopic Abundance of CO_2 Sample

We determine the abundance of the $^{12}\text{C}^{16}\text{O}_2$ isotopologue studied in this work based on measurements of the $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ isotopic ratios using an isotope ratio mass spectrometer at the CU Boulder INSTAAR Stable Isotopes Laboratory. These measurements return isotope ratios of $-40.6 \pm 0.1 \text{ ‰}$ and $-38.4 \pm 0.1 \text{ ‰}$ for $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ for our sample, respectively. We convert the measured isotope ratios to a fractional abundance of $^{12}\text{C}^{16}\text{O}_2$ using the method described in Ref. [68]. This results in a $^{12}\text{C}^{16}\text{O}_2$ fractional abundance of $98.4665 \pm 0.0002\%$. We correct our measured intensities to reflect this fractional abundance, although the magnitude of the correction is negligible relative to the other uncertainty sources that impact the measured intensity constraint parameters (discussed in more detail in Appendix 4).

1.5. Baseline correction

Although we normalize each measured spectrum using two different background spectra, slow baseline variation remains after this normalization process. We account for the remaining baseline contribution by fitting a single, 15th order Chebyshev polynomial during the multispectrum fitting process. After accounting for the baseline contribution, we observe minor residual variations in the baseline intensity with an amplitude $< 0.1\%$ transmission. As such, we judge the uncertainty in the baseline to be 0.1% transmission, which is consistent with other past work using the same spectrometer and a similar analysis framework [69].

1.6. Water vapor subtraction

For most of the spectra measured in this work, interfering water vapor lines are removed to the noise level of the measurement after being fit with a model employing argon-broadened spectral parameters and a qSDVP profile. In some spectra (namely at the highest temperatures), the presence of numerous hot lines results in some remaining water vapor residuals that are visible at the $\pm 0.1\%$ transmission level. Although most of these residuals are sufficiently isolated from CO_2 lines, some do overlap. Where significant overlap occurs, we do not attempt to fit the CO_2 line to extract line shape parameters. Additionally, if a water vapor residual adversely affects a measured transition but is not noticed, we assume that the retrieved parameters would exhibit increased residuals, high statistical uncertainty, and/or non-physical values and thus would be omitted based on our quality control criteria outlined in Section 3.2.

2. Tests of temperature-dependence models for γ , δ , and a_w

As discussed in Section 3.1, we model the temperature dependence of the broadening and pressure shift using a single power law (Eq. (2)). We test the single power law model across our full temperature range using the broadening and pressure shift coefficients measured individually at each temperature in the test matrix (see Section 6). For the test, we linearize Eq. (1) as $\frac{1}{n_\gamma} \log\left(\frac{\gamma(T)}{\gamma(296)}\right) = \log\left(\frac{296}{T}\right)^{n_\gamma}$, where n_γ is the measured temperature dependence of the broadening coefficient, $\gamma(296)$ is

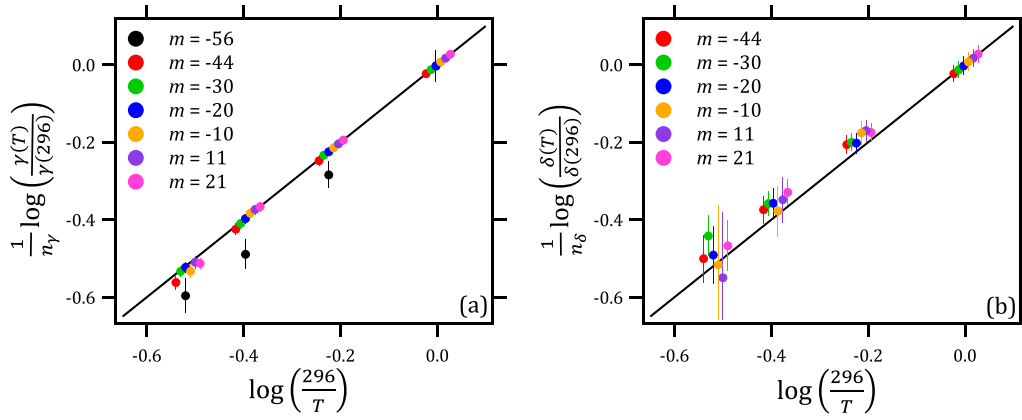


Fig. 10. Tests of the single power law temperature-dependence model for the broadening coefficient (panel (a)) and pressure shift coefficient (panel (b)) for a representative set of transitions in the 00031 $\leftarrow$ 00001 band. Data points are offset for clarity.

the broadening coefficient measured at 296 K, and $\gamma(T)$ is the broadening coefficient measured at temperature T . An analogous version is used for the pressure shift coefficient. Using this expression, we expect the measured parameters to fall on a straight line with slope one. Fig. 10 shows plots of the linearized single power law expression for the broadening and pressure shift coefficient for a representative set of transitions in the 00031 $\leftarrow$ 00001 band.

For the broadening coefficient (Fig. 10(a)), the measured parameters exhibit excellent linearity, with deviations from the expected values less than the measurement uncertainty for the majority of the transitions. Notably, the highest $|m|$ transitions ($\sim |m| > 44$) exhibit deviations greater than the uncertainty. It is unclear whether this effect is due to a bias caused by the lower SNR of these transitions or due to different temperature dependence behavior for the high $|m|$ transitions. The measured pressure shift coefficients (Fig. 10(b)) again show good linearity across the full temperature range. In this case, the observed deviations from the single power law prediction are generally on the order of the measurement uncertainty.

We use a similar approach to test for the temperature dependence of the speed-dependence parameter. As mentioned in Section 3.1, there is considerable uncertainty surrounding whether the speed-dependence parameter is temperature independent [47,50] or whether it exhibits a unique temperature dependence [14,51]. Our multispectrum fitting code assumes that the speed-dependence parameter is temperature independent. We test this assumption by measuring the speed-dependence parameter individually at each temperature in the test matrix. In this case, we only fit the strongest P-branch transitions, as the lower SNR of the high-temperature spectra prevents reliable speed-dependence retrievals for all but the strongest transitions. The measured speed-dependence parameters are shown in Fig. 11. In the bottom panel of Fig. 11, we also plot the quantity $\epsilon = |a_w(T) - a_w(298)|/\delta a_w$, where $a_w(T)$ is the speed-dependence value measured at temperature T , $a_w(298)$ is the average value at 298 K, and δa_w is the uncertainty in the measured $a_w(T)$. In this sense, ϵ is an indicator of deviations from the average speed dependence at 298 K (and therefore an indication of any measured temperature dependence). Values less than one indicate deviations that are less than the measurement uncertainty, while values greater than one indicate significant deviations. Fig. 11 indicates that the speed dependence measured at 737 K is significantly elevated relative to the average value at 298 K. At 980 K, the values are again elevated, but the deviations are on the order of the measurement uncertainty.

Notably, while these results do indicate that the speed-dependence parameter is temperature dependent, we find that the average value measured at 298 K (0.152) differs by only 2% rel-

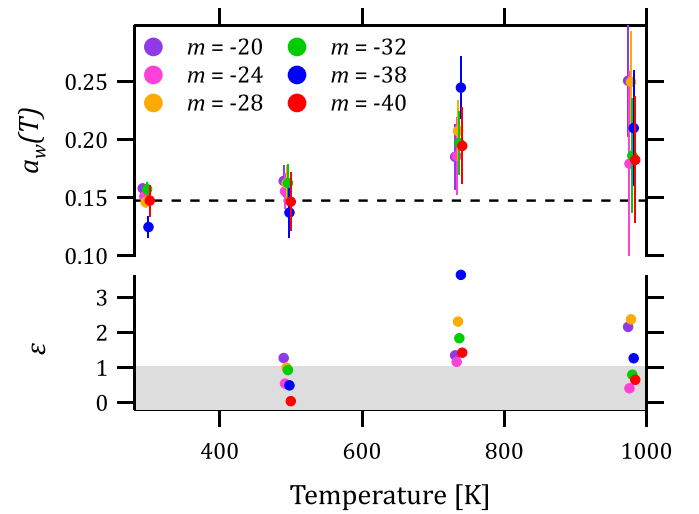


Fig. 11. Speed-dependence parameters measured at the four temperatures of the test matrix for a representative set of strong P-branch transitions in the 00031 $\leftarrow$ 00001 band. The dashed line indicates the average speed dependence measured at 298 K. The bottom panel plots the quantity ϵ (see text for definition), which indicates whether observed deviations from the average at 298 K are greater than ($\epsilon > 1$) or less than ($\epsilon < 1$) the measurement uncertainty. Data points are offset for clarity.

ative to the average value (0.155) measured using the full, multispectrum fitting approach that includes all temperatures in the test matrix. The agreement between these two values indicates that our measured speed-dependence parameters are biased by less than 2% (on average) by the assumption that the speed dependence does not depend on temperature. We attribute this fact to the strongest speed-dependence residuals being present in the spectra measured at 298 K (owing to the higher SNR). This effectively constrains the measured speed-dependence values in the multispectrum fit solution to be very close to their values at ambient temperature. Based on this finding, we include a 2% relative uncertainty term in our estimate of the total uncertainty in the measured speed-dependence parameters (discussed in more detail below).

3. Effect of line mixing on retrieved spectral parameters

The omission of line mixing in our spectral model could potentially affect the measured line shape parameters as the fit solution may try to compensate for line mixing effects through changes

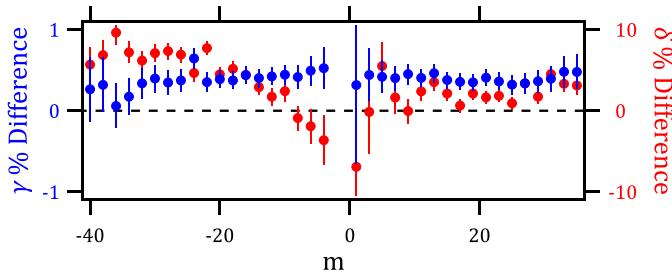


Fig. 12. Percent difference in the broadening coefficient (blue) and pressure shift (red) after the addition of first order line mixing coefficients. We estimate the total effect of line mixing on the two parameters as the average magnitude of the percent difference.

to the Voigt or qSDVP parameters [17]. However, the multispectrum fitting approach adds some degree of robustness against this effect because the retrieved parameters are constrained in the solution by the low-pressure and high-temperature spectra that are minimally influenced by line mixing. We estimate the effect of line mixing on our measured spectral parameters by refitting the spectra measured at 296 K with and without the first-order (Rosenkranz [70]) line mixing parameters calculated by Hashemi et al. [14]. Fig. 12 shows the percent change in the measured broadening coefficients and pressure shifts after the addition of the first-order line mixing coefficients to the Voigt profile. We estimate the total effect of line mixing on each parameter as the average magnitude of the percent difference. We find that the first order line mixing coefficients change the retrieved parameters by an average of $\sim 0.39\%$ and 3.8% for the broadening coefficient and the pressure shift coefficient, respectively. We add these values in quadrature with the other uncertainty sources (discussed below) to estimate the total uncertainty in the measured parameters. A full study of line mixing at high temperature, including both these and higher density spectra, will be the subject of an upcoming paper.

4. Estimation of line shape parameter uncertainties

4.1. Line position

We constrain the line positions of all lines in a given vibrational band to follow a simple offset (Δ_σ) from the line positions listed in the HITRAN2016 database (see Section 3.2). We estimate that the uncertainty in Δ_σ is impacted by the uncertainty in the spectrometer frequency axis and the statistical uncertainty reported by the fitting algorithm. We add these two components in quadrature to assess the total uncertainty in the line position constraint parameter.

4.2. Intensity

Similar to the line positions, we constrain the intensities of all lines in a vibrational band using a global scaling factor relative to the database intensities. In this case, the uncertainty in the intensity constraint parameter depends on the statistical uncertainty reported by the fitting algorithm and the uncertainty in the column density of the gas. We assume the column density is a function of the gas cell path length (L), the abundance of the principle CO_2 isotopologue (A), the temperature (T), and the pressure (P), so that the fractional uncertainty in the column density (N_c) is given by

$$\frac{\delta_{N_c}}{N_c} = \sqrt{\left(\frac{\delta_L}{L}\right)^2 + \left(\frac{\delta_A}{A}\right)^2 + \left(\frac{\delta_T}{T}\right)_{\max}^2 + \left(\frac{\delta_P}{P}\right)^2} \quad (12)$$

We add the output of Eq. (12) in quadrature with the statistical uncertainty to estimate the total uncertainty in Δ_σ .

4.3. Broadening coefficient and speed-dependence parameter

The total uncertainty in the measured broadening coefficients is a combination of the statistical uncertainty reported by the multispectrum fitting algorithm, the estimated uncertainty due to the omission of line mixing in the spectral model (assessed in Appendix 3), and uncertainty contributions due to the experimental pressure and temperature. Following Ref. [71], we assume that the relative uncertainty in the broadening coefficients due to uncertainty in the experimental gas pressure is that of the pressure sensor (0.25%). Additionally, we quantify the impact of the temperature uncertainty on the broadening coefficients using the relation [71]

$$\delta_{\gamma_T} = \gamma \left[\left(\frac{T + \delta_T}{T} \right)^{n_\gamma} - 1 \right] \quad (13)$$

where δ_{γ_T} is the uncertainty in the measured broadening coefficient γ due to an experimental temperature uncertainty δ_T at temperature T . Following [71], we assume a worst-case value of $n_\gamma = 1$ to estimate the uncertainties. Here, it is important to note that Eq. (13) is not strictly valid for the multispectrum analysis used in this study since multispectrum fitting interrogates multiple temperatures at once. As such, the uncertainties estimated using Eq. (13) constitute an upper limit to the uncertainty in the broadening coefficient due to temperature uncertainty. To apply Eq. (13) to the multispectrum fit results, we assume the worst-case temperature uncertainty of ± 12.7 K at 980 K.

We assess the total uncertainty in the measured broadening coefficients by adding all uncertainty contributions in quadrature. Additionally, we assume the speed-dependence parameter is impacted by the same uncertainty contributions, and use the same approach to assess the total uncertainty in the speed dependence. In this case, we also include a 2% relative uncertainty term due to the un-modeled temperature dependence of the speed dependence (see Appendix 2).

4.4. Temperature dependence of the broadening coefficient

As discussed in Refs. [33,72,73], the statistical uncertainties of the measured temperature dependence exponents are highly correlated with the uncertainties in the broadening coefficients, meaning the temperature dependence exponents for the broadening coefficients cannot be known better than the broadening coefficients themselves. We account for this fact using the expression derived in Refs. [72,73] that provides a lower limit for the uncertainty of measured temperature dependence exponents. The limit is given by

$$\delta_{n_\gamma} > \sqrt{2} \left(\frac{\delta_\gamma}{\gamma} \right) \ln \left(\frac{T_{\max}}{T_{\min}} \right)^{-1} \quad (14)$$

where δ_{n_γ} is the absolute uncertainty in the temperature exponent, δ_γ/γ is the relative uncertainty in the broadening coefficient, and $T_{\max}$ and $T_{\min}$ are the maximum and minimum temperatures used in the fit to determine the temperature dependence exponents.

We estimate the total uncertainty in the temperature dependence exponents for the broadening coefficient by adding the statistical uncertainty reported by the multispectrum fitting algorithm in quadrature with the uncertainty component calculated using Eq. (14).

4.5. Pressure shift coefficient and its temperature dependence

The pressure shift coefficient and its temperature dependence are similar to the broadening coefficient in that the pressure shift

coefficient is also influenced by the omission of line mixing and is linear with the experimental pressure. Further, its temperature dependence is assumed to follow the same power law form used to model the temperature dependence of the broadening coefficient. As such, we assume the total uncertainty in the pressure shift coefficient is influenced by the same factors as the broadening coefficient, and we estimate the uncertainty in the pressure shift coefficient and its temperature dependence using the same procedure outlined above for the broadening coefficient.

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